

1 Problem Description

Between 0 and 100 hours (not including 100:00:00) on stopwatch how many times will exactly 2 unique digits be displayed.

2 Problem Solution

The Digits ranges are like so:

$$Digit_{H,0} : 0 - 9$$

$$Digit_{H,1} : 0 - 9$$

$$Digit_{M,0} : 0 - 5$$

$$Digit_{M,1} : 0 - 9$$

$$Digit_{S,0} : 0 - 5$$

$$Digit_{S,1} : 0 - 9$$

So there are 4 between 0 and 9 AND there are 2 between 0 and 5.

For the combinations where the digits are less than 6 there are $2^6 \times \binom{6}{2} = 64 \times 15 = 960$ combinations. This is because for a unique pair, each digit can be 1 of 2 states, so there are 2^6 combinations. And there are 15 unique pairs.

For the combinations where one of the digits is ≥ 6 and the other is ≤ 6 there are $2^4 \times 4 \times 6 = 384$ combinations because for the digits between 6 and 9 there are 4 with 4 different numbers and for the digits between 0 and 5 there are 2 with 6 different numbers however these 2 must be the same.

For the combination where both of the digits are ≥ 6 then there are $2^6 \times \binom{4}{2} = 64 \times 6 = 384$ combinations.

So in total there are $960 + 384 + 384 = 1728$ combinations